Where Does Noise Robustness Live? Evidence from Ablation, Role Swaps, and Cross-Lingual Transfer in Code-Mixed Banking NLU

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Abstract

Conversational banking assistants on messaging platforms face language that their natural-language-understanding (NLU) pipelines were never designed for: vowel-dropped shorthand (*blnc chk krdo*), romanized code-mixing, colloquial verb frames. Layered pipelines stack repair rules, learned retrieval, and an LLM, but which layer carries the noise robustness is rarely measured. We ablate, replace, and transfer the defenses of a 13-layer Hinglish banking NLU (163 intents) on BankStress-330, a released three-tier diagnostic stress set, $\geq$3 repeats per configuration: LLM-in-the-loop configurations flip up to $\sim$3% of verdicts across identical temperature-0 runs, defeating single-run ablation. A 2$\times$2 factorial shows the Hinglish normalizer helps a base encoder but is redundant after noisy-query fine-tuning; the post-fine-tuning redundancy also holds on public Hinglish-TOP (n=6,513, McNemar z=5.0, $p \approx 5\times10^{-7}$). Only removing the conditional LLM arbiter collapses adversarial accuracy (.739$\rightarrow$.545); neither tested substitute matched it, yet the same fine-tuned cross-encoder raises every tier when restricted to reranking (directional, unresolved at n=330). On native-reviewed Tamil and Telugu probes normalization is unstable in sign. Robustness appears to migrate into tuned components, only for the pair tuned on.

1 Introduction

WhatsApp-register Hindi–English (“Hinglish”) banking queries combine four noise sources: orthographic noise (typos, vowel-dropped SMS shorthand), code-mixing (Latin-script Hindi with embedded English banking nouns), colloquial lexical variety (*karza nipta do* for loan foreclosure), and underspecification against a fine-grained intent taxonomy. Pretrained encoders degrade substantially on such input (Kumar et al., 2020; Aliakbarzadeh et al., 2025), and clean or translated test sets overestimate real performance on native noisy queries (He et al., 2026). Production systems answer with layered pipelines: deterministic normalization and rules up front, learned retrieval in the middle, an LLM at the end. Each layer was added for a reason, usually a remembered failure. Whether each layer still *earns its place* is a question deployed systems rarely revisit.

The question is live because the literature answers it both ways. Lexical normalization improved downstream tagging in code-switched settings as recently as 2021 (van der Goot and Çetinoğlu, 2021; van der Goot et al., 2021), yet noise-pretrained or noise-fine-tuned models have repeatedly made input cleanup redundant (Nguyen et al., 2020; Zhuang and Zuccon, 2022; Maarouf and Tanguy, 2025). Contemporary code-mixed fintech assistants nonetheless still classify on normalized queries (Hazarika et al., 2025).

We revisit the question on a 13-layer NLU built for a WhatsApp banking assistant. The system is a single-team research build; it does not serve live bank customers, and we claim no production deployment. By *production-style* we mean, operationally: enforced latency budgets (median hot-path $\leq$120 ms), an open-weight-only on-premise constraint, a 163-intent taxonomy, and a compliance-driven personally-identifiable-information (PII) layer: constraints that shaped the architecture and that a research prototype would not face. Our method is deliberately simple: remove one layer at a time, substitute candidate replacements, re-measure on a tiered benchmark; then repeat the key configurations on a language pair the pipeline was never tuned for, and replicate the central finding on the public Hinglish-TOP benchmark (n=6,513). We do not claim generality across architectures; we claim the *question* is answerable by measurement, and report the answer for this pipeline. Contributions:

- **C1: Tiered evaluation protocol.** A repeatable measurement protocol (tier-stratified stress queries, 330 over 22 core intents classified against the full 163-intent space; $\geq$3 repeats per configuration with variance reported; paired per-query tests), released as the BankStress-330 benchmark with per-query authoring notes and the full adjudication record. The protocol also yields a measured $\sim$3-point single-run verdict noise floor for LLM-in-the-loop ablation. We are not aware of a public Hinglish banking intent benchmark, despite public code-switched and banking benchmarks in adjacent spaces (Khanuja et al., 2020; Aguilar et al., 2020; Casanueva et al., 2020; Agarwal et al., 2023).

- **C2: Factorial layer ablation.** A factorial experiment showing hand-built normalization helps a base encoder but becomes redundant once the encoder is fine-tuned on noisy queries; the rules engine is a latency feature; only the bounded LLM arbiter is load-bearing among the ablated layers under the substitutions we test.

- **C3: Replacement and escalation study.** The same fine-tuned cross-encoder fails as a *decider* (.461) and wins as a *ranker* (.906), a mechanism-supported result (shortlist recall, escalation reduction); its end-to-end accuracy delta is directional at our benchmark size. An 8B arbiter recovers only a fraction of the 20B's value; failure-targeted augmentation shows dose-dependent harm. Unlike C2, these findings rest on the in-house benchmark alone (§3). The escalation trigger tracks noise tier without labels (67/73/94%), but its *rescue rate* is inverted ($\sim$26% of adversarial vs. $\sim$13% of clean escalations).

- **C4: Transfer probes.** Across three native-validated unseen-pair sets (two independent Tamil–English translators; Telugu–English), the normalizer's contribution is unstable in sign (-3.3, -2.5, $+2.9$): off the tuned pair, no layer's role can be assumed.

Read together these are one argument: C1 is the guardrail that makes layer attribution trustworthy; C2 locates robustness in functional roles, not rules or scale; C3 is its sharpest instance; C4 bounds it to the tuned pair.

2 System

The pipeline (Fig. 1) processes each utterance through: (L1) PII masking with checksum validation (Luhn for cards, Verhoeff for national IDs); (L2) a 10-stage Hinglish normalizer ($\sim$80 rule families: vowel-restoration regexes such as *kr*→kar**, Indian numeric grammar such as *lakh/crore* and fraction words *sawa/paune*; with PII spans frozen as placeholders); (L3) language identification (LID; two-stage FastText with a banking-loanword suppression list); (L4) sentence encoding with multilingual-e5-base plus a low-rank-adaptation (LoRA) adapter fine-tuned on 4,300 hard-negative triplets and 1,600 noisy queries; (L5) centroid-based domain routing; (L6) a deterministic rules engine (fast exit at confidence $\geq$0.95); (L7) per-domain hybrid retrieval (dense FAISS nearest-neighbor search + lexical BM25, reciprocal-rank fusion, lexical weight down-scaled for stopword-heavy queries); (L8) score-cliff detection; (L9) a 7-signal out-of-scope ensemble (cf. Larson et al., 2019); (L10) a decision matrix emitting EXECUTE/RERANK/CLARIFY/OOS; (L11) a *conditional* LLM arbiter (gpt-oss-20b, an open-weight model prompted to pick among retrieved candidates), invoked only when the decision matrix cannot certify a clear winner; (L12) risk-aware confidence adjustment; (L13) slot extraction.

Two design properties matter for what follows. The pipeline *abstains into arbitration*: the escalation decision is deterministic and precedes any LLM call. And the arbiter is bounded: it selects among retrieved candidates rather than free-generating, and its output passes through downstream confidence gates.

3 Experimental setup

Benchmark. We construct and release **BankStress-330**, a diagnostic stress set for layer attribution (not a leaderboard benchmark): 330 queries over 22 core banking intents: CLEAN (66; canonical phrasings), MESSY (154; typos, shorthand, code-mix), ADVERSARIAL (110; colloquial idioms, implicature, zero domain keywords). Each query is classified against the full 163-intent space. Because the benchmark was authored by the same author who built the system, its notion of "hard" co-evolved with the pipeline; of our four findings, only the normalizer result is additionally replicated on public data; the arbiter and reranker

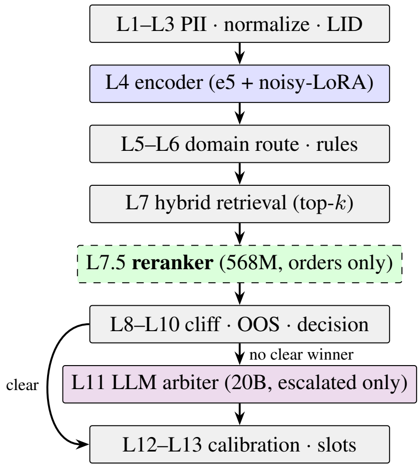

Figure 1: The 13-layer pipeline (condensed). Gray layers are deterministic; blue is the learned encoder; green (dashed) is the reranker intervention evaluated in §4.4; violet is the conditional LLM arbiter.

findings rest on this benchmark alone and are scoped accordingly; conversely, every structural claim with a public vehicle is checked on public data (§4.1). The set is deliberately confusion-heavy; escalation rates on it are stress-set figures (production-representative traffic, i.e., internally generated test conversations replaying realistic usage, no real customers, escalates $\sim$5% of turns, §6), not traffic estimates.

Protocol. Every LLM-bearing configuration runs $\geq$3$\times$; deterministic configurations are reported once after verifying bit-exact outputs. All runs use one machine (CPU inference for the pipeline; the LLM arbiter served via API at temperature 0; all models in the stack are open-weight and on-premise deployable, a banking compliance requirement). We report per-tier accuracy as mean with min–max across repeats, exact McNemar on majority-vote failure sets, and paired bootstrap CIs over per-query correctness. Ablations stub exactly one layer; the reranker experiment inserts exactly one component between L7 and L8.

Reproducibility observation. Configurations without the LLM arbiter reproduce *bit-identically* across repeats run under identical harness code;[1] configurations containing it flip 1–3% of per-query verdicts between identical temperature-0 runs (3–9 per 330 across all 21 pairings of 7 released baseline repeats, mean $\sim$2%; 15/330 queries flip in at least one repeat). The deterministic layers are exactly reproducible; the LLM contributes all run-to-run variance. This has a methodological implication beyond our study: any *single-run* ablation of an LLM-in-the-loop pipeline carries a $\sim$3-point verdict noise floor, so differences below that require repeated-measures testing, which is why every comparison in this paper uses $\geq$3 repeats for LLM-bearing configurations, while deterministic configurations (including the public-data runs) are single-run bit-exact under fixed code and weights; we report three decimals for traceability but do not interpret differences within this noise floor. A run ledger maps every table to its result files, run counts, and harness era (`MANIFEST.md` in the artifacts). We do not correct for multiple comparisons across the many configurations tested: the load-bearing contrasts survive any standard correction by orders of magnitude, and the borderline ones (reranker, transfer) are interpreted as directional precisely because they would not.

Config	CLN	MSY	ADV	Overall
Full pipeline	.985	.948	.739	.886
– Normalizer	.980	.961	.752	.895
– Rules engine	.975	.951	.739	.885
– LLM arbiter	.924	.844	**.545**	.761

Table 1: Layer ablation (mean of 3 repeats; ranges in text); higher-without-layer rows mean the layer was net-neutral or harmful post-fine-tuning. Median (p50) latency: $\sim$440 ms with the arbiter, $\sim$60 ms without.

4 Results

4.1 Which layers carry noise robustness

On our benchmark, removing the hand-built normalizer does not reduce accuracy (Table 1; 7 fixed vs. 7 broken discordant queries on majority-vote failure sets over the released repeats; p=1.0): the rules occasionally rewrite away signal: removing the normalizer *fixes* a typo query (*mera **crd** card chori ho gya*). The rules engine is a latency feature: accuracy is unchanged without it. Only the arbiter's removal collapses the adversarial tier (-19.4 points; 51 queries saved vs. 6 cost on majority-vote failure sets over the released repeats; p=5.7×10^{-10}).

A factorial experiment isolates the cause (Table 2). We rebuilt the pipeline's indices with the *base* encoder (no noisy-query LoRA) and crossed encoder training with the normalizer. Before noisy

[1] A mid-study harness patch (adding per-query logging and ablation stubs) shifted exactly one –arbiter query (.761→.758); both eras' files are released and labeled.

Encoder	Norm. ON	Norm. OFF	Norm. effect
Base	.857	.833	+2.4 (+5.5 **adv**)
Noisy-FT	.886	.895	−0.9 (n.s.)

Table 2: Factorial: encoder training × normalizer (overall accuracy; 3 repeats/cell). The normalizer helps the base encoder and is redundant after noisy fine-tuning.

fine-tuning the normalizer genuinely helps, consistent with positive extrinsic results in the normalization literature (van der Goot and Çetinoğlu, 2021; van der Goot et al., 2021). After fine-tuning on 1,600 noisy queries its benefit disappears. The interaction supports the reading that noisy fine-tuning made normalization redundant in the tested settings (cf. Nguyen et al., 2020; Zhuang and Zuccon, 2022). The fine-tune itself is worth +2.9 to +6.2 points depending on the normalizer condition. **The post-fine-tuning redundancy replicates on public data** (the base-encoder benefit does not: on Hinglish-TOP the normalizer has no effect even untuned, consistent with domain-matched rules being its precondition). We rebuilt the pipeline's taxonomy and indices from the Hinglish-TOP training split (Agarwal et al., 2023) (58 intents; reminders, weather, media; no banking) and evaluated its 6,513-query human-annotated test set (105 duplicate surface forms in the public split are retained as distinct examples) in the deterministic no-arbiter configuration (single runs are bit-exact). With the noisy-query LoRA, removing the normalizer *improves* accuracy .617→.631 (fixes 202 queries, breaks 112; McNemar z=5.0, $p \ll 0.001$); with the base encoder the normalizer has no effect (.603 both). Two refinements follow: the encoder's noise robustness transfers cross-domain (+1.4 to +2.7 over base), and the normalizer only ever helped when the encoder was untuned *and* its rules were domain-matched (our in-house base-encoder cell). We use *migrate* operationally: a layer's accuracy contribution becomes indistinguishable from zero while the encoder's grows, and reappears on a pair the encoder was not tuned for (§4.5).

4.2 Escalation rises with difficulty; rescue efficiency stays low

The decision matrix escalates 67% of CLEAN, 73% of MESSY, and 94% of ADVERSARIAL queries; escalation rate tracks tier difficulty with no difficulty labels anywhere in the system. But comparing baseline against the –arbiter configuration reveals the arbiter's *rescue rate* (measured indirectly: the per-query outcome delta between the two configurations under a rest-of-pipeline invariance assumption): the arbiter turns a wrong answer into a correct one for ∼26% of escalated adversarial queries (24.3–27.2 across 16 pairings of independent baseline and –arbiter repeats), ∼16% of messy (15.9 across all pairings), and only ∼13% of clean ones (11.4–13.6); the reverse (harm) direction is what the paired repeats' spread reflects, and on public data the two directions cancel (§4.3). Escalation *rates* carry no such interval: the trigger is upstream of the LLM and reproduces exactly. Most clean escalations are risk-averse confirmation (confusion-group policy), not necessity: 44 clean-tier LLM calls yield ∼6 rescues against ∼1 harm; in-domain harm stays at 1–4% of escalations on every tier, whereas off-domain rescue and harm cancel (§4.3). Per-query latency splits at ∼70 ms (hot path in the baseline configuration; Table 1's ∼60 ms is the –arbiter configuration's overall p50) vs. 417–530 ms (escalated). Figure 2 visualizes the inversion.

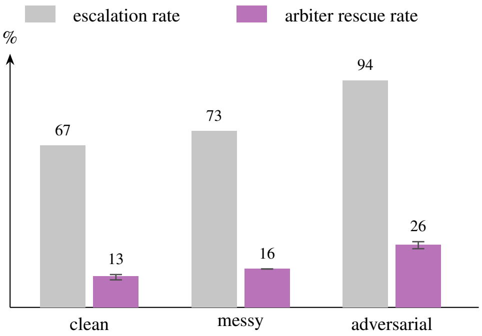

Figure 2: Escalation rate tracks noise tier, but the arbiter's *rescue rate* (share of escalations where it turns a wrong answer into a correct one) is lowest exactly where escalation is cheapest to skip. Whiskers: min–max across 16 pairings of independent baseline and –arbiter repeats; rates are deterministic and carry no interval.

The wasted-escalation problem has begun to be named in the cascade literature (Jitkrittum et al., 2023; Rabanser et al., 2025; Bouchard, 2026); stratifying escalation and rescue rates by input noise tier is, to our knowledge, new, and it identifies exactly where a cheaper intermediate stage would pay, which §4.4 exploits.

4.3 What does not replace the arbiter

A 568M multilingual cross-encoder (bge-reranker-v2-m3, a reranking fine-tune released along-

Attempt	Overall	ADV
Arbiter substitutes		
Zero-shot xenc as decider	.530	.336
Fine-tuned xenc as decider	.461	.282
8B instruct LLM as arbiter	.810	.579
Augmentation (contrast)		
Full dose	.873	.758
Half dose (holdout eval)	.893	.757

Table 3: Interventions that did not pay off. Rows 1–3 substitute the arbiter; rows 4–5 are augmentation (a different intervention, shown for contrast). Note the half-dose row exceeds baseline overall yet fixes only 2/21 held-out failures: targeted gains, no generalization.

Config	MSY	ADV	Overall	Esc.
Baseline	.948	.739	.886	260
+ zero-shot rerank	.929	.779	.890	210
+ FT rerank	**.961**	**.782**	**.906**	242

Table 4: Reorder-only reranking (α=0.3 blend; mean of three designated table runs, see `MANIFEST.md`; CLEAN is .985 throughout). Esc. = LLM escalations per 330. Only the fine-tuned reranker raises every tier's mean; the zero-shot variant trades MESSY for ADVERSARIAL.

side M3-Embedding; Chen et al., 2024; Beijing Academy of Artificial Intelligence (BAAI), 2024) substituted for the arbiter fails zero-shot (colloquial-Hinglish-to-intent matching is out-of-distribution for passage-relevance training) and fails *worse* after fine-tuning on 21k taxonomy pairs: it learns to recognize familiar phrasings, but escalated queries are precisely the unfamiliar ones (Table 3). Rerankers also emit relative orderings, not calibrated confidences; forcing their scores through absolute-confidence gates misuses them (Yan et al., 2022). A smaller *decoder* fails too: swapping the 20B arbiter (gpt-oss-20b) for an 8B instruct model (Llama-3.1-8B-Instruct; same prompt, same role; 3 repeats) recovers only a fraction of its value: adversarial .573–.591 across repeats (the table reports the mean), barely above running *no* arbiter (.545) and far below the 20B's .739. We report this as a model-swap observation (none .545 $\rightarrow$ 8B $\sim$.58 $\rightarrow$ 20B .739): the two arbiters differ in training data and instruction tuning, not only in size, so we cannot attribute the gap to parameter count alone. The arbiter's value is also domain-bound: on public Hinglish-TOP (non-banking, 58 intents), enabling it over the deterministic configuration is a wash (+0.17pp, n=6,513; 959 queries fixed vs. 948 broken, exact McNemar p=0.82; 76% invocation). It recovers 39.9% of the deterministic failures while breaking a nearly equal number of successes (959 fixed vs. 948 broken), so its large in-domain value (Table 1) does not transfer off the banking distribution; run files are released.

The augmentation rows document a known risk (He et al., 2023; Joshi and He, 2022) materializing, with a dose–response. Generating colloquial phrases from observed failures (practice: Wei and Zou, 2019; Sahu et al., 2022, 2023) at full dose (1,303 accepted phrases) shifts intent centroids enough to break the easy head (CLEAN −4.1); accumulation alongside real data, the standard remedy in the synthetic-data literature (Gerstgrasser et al., 2024), did not prevent it. At half dose the head survives intact, but only 2 of 21 *held-out* failures are fixed. Failure-targeted augmentation fixes what it targets, barely generalizes to untargeted failures, and harms broadly only as the dose grows (cf. Li et al., 2023; Ma and Zhang, 2026).

4.4 What helps: the same model, in its proper role

Inserting the *fine-tuned* cross-encoder between retrieval and score-cliff (rescoring the top-8, blending with retrieval scores, reordering, deciding nothing) measurably improves what the arbiter sees: shortlist recall@1 rises from .701 to .736 and recall@3 from .874 to .897 on escalated queries, and adversarial escalations fall from 94% to 85% while adversarial accuracy *rises* (Table 4). The same weights yield an end-to-end pipeline at .461 when substituted as the decider and at .906 when restricted to ranking. Statistically: across every released repeat of both arms (15 reranked, 16 baseline; all runs reported, none dropped), reranked runs match or exceed baseline runs in 231 of 240 pairwise comparisons; the 9 exceptions all pair the two highest baseline repeats with the three lowest, earliest-harness-era reranked runs. The paired per-query bootstrap gives +1.6 overall with 95% CI $[0.0, +3.3]$; the lower bound sits at zero. The coarser majority-vote McNemar is non-significant (p=0.18). We therefore claim a consistent directional gain that n=330 cannot resolve to significance, and rest the finding on the mechanism measurements rather than on the accuracy delta. Hyperparameter provenance: the blend weight was chosen from a two-value probe ($\alpha \in \{0.5, 0.3\}$, single runs) on this same benchmark, and the

Config	Overall	p50 (ms)	LLM calls
– LLM arbiter	.761	∼60	0
8B arbiter	.810	∼405	260
Baseline (20B)	.886	∼435	260
+ zero-shot rerank	.890	∼425	210
+ FT rerank	**.906**	∼460	**242**

Table 5: Accuracy–cost frontier across configurations (stress set).

	ta-T1 (80)	ta-T2 genz (80)	te (40)
Baseline	.800	.825	.742
– Normalizer	.767 ↓	.854 ↑	.717 ↓
– LLM arbiter	.762	–	.625
+ FT reranker	.788 ↓	–	.708 ↓

Table 6: Transfer probes: two independent Tamil–English translations (standard register T1; Gen-Z register T2) and Telugu–English, all native-validated, zero-shot (no Tamil or Telugu anywhere in training). Arrows: direction of change vs. each set's own baseline. Empty cells: configurations not run on T2, where the compute budget prioritized the normalizer contrast; the reranker and arbiter transfer claims therefore rest on T1 and Telugu only.

shortlist depth is the retriever's fixed top-8; no disjoint tuning split existed at this scale, so the +1.6 directional estimate should additionally be read as optimistic. In this cross-encoder experiment, role restriction, not model scale, determined whether it helped or harmed the pipeline, consistent with ranking theory: learning-to-rank objectives leave scores scale-uncalibrated, so relative order is trustworthy where absolute thresholds are not (Yan et al., 2022; Yu et al., 2025; Baral et al., 2026). Prior pairwise intent deciders succeeded in *clean few-shot* settings (Zhang et al., 2020); in our high-label-space, colloquial, escalated-query regime, the decider role fails while shortlist reranking succeeds. The closest system-level prior routes between two deciders on uncertainty (Arora et al., 2024); restructuring the arbiter's candidate list is a different, and here better, use of the small model. Table 5 summarizes the accuracy–cost frontier: the fine-tuned reranker improves accuracy while reducing LLM calls. Its measured p50 does not establish an end-to-end speedup: the reranker adds local compute, LLM-bearing p50 is API-dominated and varied across runs, and the FT-rerank p50 is if anything higher than baseline. The meaningful cost contrast is LLM calls, not latency. The no-arbiter configuration is a separate frontier point, viable wherever 12 points of accuracy can be traded for the removed API round-trip.

4.5 Transfer: off-distribution, layer roles become unstable

Romanized Dravidian–English code-mix has its own literature and shared tasks (Chakravarthi et al., 2022); intent-labeled resources, however, remain scarce. We had a native Tamil speaker translate and validate a stratified 80-query slice (20/30/30 across tiers) into Tanglish, romanized Tamil–English code-mix as a Tamil customer would type it. The pipeline has *no* Tamil training anywhere: the normalizer is Hinglish-tuned, the LoRA saw only Hinglish noise, the reranker was fine-tuned on the Hinglish taxonomy. Four observations (Table 6). First, zero-shot transfer is viable on every set (.74–.83): multilingual encoding plus English banking loanwords carry most of the signal. Second, and centrally: **the normalizer's contribution off-distribution is unstable in sign.** It helps on standard-register Tanglish (-3.3 when removed; paired bootstrap 95% CI $[-8.1, +0.9]$; all three –normalizer repeats fall below all three baseline repeats, but, like the reranker effect, the effect is not resolved to significance at n=80) and on Telugu (-2.5 in the mean; the Telugu repeats overlap the baseline range, so this set supports mean-consistency only), but, as observation three shows, not universally. A stage-wise ablation on the Tanglish set attributes $+2.1$ of the benefit to unambiguously language-agnostic stages and $+1.2$ to the strictly Hindi-specific stages, jointly accounting for the full 3.3-point gap. Third, **the direction reverses on the Gen-Z-register Tamil set** ($+2.9$ when removed, with all three –normalizer repeats *above* all three baseline repeats; the T1/T2 sign contrast, opposite and repeat-consistent on both sides, is the robust observation): the normalizer's sign is unstable across two independent renderings of the same content. Because each register has a different translator, register and translator idiosyncrasy are confounded (Limitations) and we do not attribute the reversal to register alone; either reading warns that single-translator transfer evaluations can mislead in both direction and magnitude. Fourth, two effects are stable across every set where tested: the Hinglish-fine-tuned reranker improves neither tested pair and the LLM arbiter remains load-bearing (Telugu: -11.7 without it); both were tested on T1 and Telugu only, not on T2. We state the scope plainly: n=40–80 per set, one translator per set, a loanword-heavy domain that likely flatters transfer. The safe architecture lesson is accordingly conservative: off-distribution, no layer's contribution can be assumed: it must be measured per pair and register. Zero-shot transfer stayed viable everywhere (.74–.83), consistent with the encoder carrying the transferable signal; every other layer was pair-dependent, and even the arbiter is a wash on public non-banking data (§4.3).

5 What remains broken

If robustness now lives in learned components and a bounded arbiter, what still defeats all of them? The residual-failure taxonomy uses majority voting (fail in $\geq$ half the runs) over four baseline repeats (identified in the artifact manifest as the set whose majority-failure set reproduces the adjudication record): 38 majority failures, with 43 failing in at least one repeat. These decompose into: **(F1) colloquial lexical variety** ($\sim$16; *kitna bacha mera*, *4 digit wala code bhool gya*; nothing misspelled, so no normalizer can help; retrieval lacks the verb frames); **(F2) pragmatic implicature** ($\sim$6; *credit card chhota pad rha h* = limit increase; the arbiter's remaining ceiling); **(F3) sibling-intent granularity** ($\sim$14; statement$\leftrightarrow$balance$\leftrightarrow$outstanding; a human would ask a clarifying question; forcing argmax is the error); **(F4) multi-question utterances**; **(F5) context-dependent fragments** (*baaki kya hai*), strictly unanswerable in a stateless evaluation. These are largely not addressable by orthographic normalization alone. An oracle analysis quantifies F3: 12/38 failures are wrong-intent-but-same-domain; converting them to clarification questions lifts the ceiling from .886 to .921, and only 5 of the 12 fall inside the system's *declared* confusion groups, i.e., the hand-maintained ambiguity configuration under-covers real ambiguity by more than half. All 38 majority-vote failure labels were re-adjudicated by the benchmark author: 35 were confirmed and 3 overturned (e.g., *Show my debit card details*, gold *dc_explore*, adjudicated to *dc_card_details*). System predictions were visible during adjudication, which risks anchoring; we release the full record so the adjudication can be independently repeated. Adjudication shifts headline accuracies by at most +0.7 points (baseline .886$\rightarrow$.893; reranked .899$\rightarrow$.902) and changes no conclusion; we report raw numbers throughout and release the adjudication record with the benchmark.

6 Evolution: measurement-driven design

The pipeline reached its current shape through five measurement$\rightarrow$defense iterations; e.g., romanized-Hinglish LID accuracy collapsed to 6.3% under banking loanwords in a development-time measurement (whose logs predate this study and are not part of the release) before a loanword-suppression list fixed it, and LLM invocation fell from 20–30% of turns to a conditional $\sim$5% on production-representative traffic (the 330-query stress set is deliberately confusion-heavy, hence its 67–94% escalation rates in §4.2). This paper is the sixth iteration of that loop, pointed at the pipeline itself.

7 Discussion

Where robustness lives. Not in the rules: three generations of repair-rule writing were made redundant *for the tuned pair* by one fine-tune on noisy queries (isolated by the factorial in §4.1). It lives in (a) the noisy-text encoder, (b) a task-fine-tuned ranker over its candidates, (c) a bounded LLM arbiter for the residue; and, for language pairs outside the fine-tuning distribution, (d) in a mixture that must be re-measured per pair and register: our deterministic scaffolding helped on two unseen sets and hurt on a third. The residual gap is lexical variety and taxonomy granularity, which argue for clarification actions and colloquial coverage, not more repair rules.

The arbiter was API-served in these experiments, though every model in the stack is open-weight and on-premise deployable. The full-dose augmentation result is in-loop by construction (the half-dose holdout addresses this). Stress-set escalation rates do not reflect production traffic distribution.

8 Related work

Concurrent submission. A concurrent, separately-submitted workshop paper by the same author studies conversational language identification for a related banking-assistant setting; its contribution (a conversational LID benchmark and a cascade system) and its data are disjoint from this paper's layer-attribution study, and no results are shared between the two. (Also declared to the program committee via the submission form.)

Lexical normalization. The W-NUT Multi-LexNorm shared task established multilingual lexical normalization and reported *positive* extrinsic effects on parsing and tagging (van der Goot et al., 2021; Samuel and Straka, 2021), with normalization of code-switched text yielding a 5.4% relative POS improvement (van der Goot and Çetinoğlu, 2021). Our results do not contradict these findings; they date them. With encoders pretrained or fine-tuned on noisy text, the picture inverts: dictionary normalization stops helping a tweet-pretrained LM (Nguyen et al., 2020), LLM-based normalization slightly *hurts* downstream sequence labeling on noisy technical text (Maarouf and Tanguy, 2025), and typo-robust *training* outperforms input cleanup in dense retrieval (Zhuang and Zuccon, 2022). We re-run this question where it has not been asked, a production code-mixed intent pipeline, and find the 2021 conclusion no longer holds there (§4.1), except on language pairs outside the fine-tuning distribution (§4.5); notably, a contemporary Hinglish fintech assistant still classifies on normalized queries (Hazarika et al., 2025).

Code-mixed NLU. Benchmarks for Hindi–English code-switching contain no intent-classification task (Khanuja et al., 2020; Aguilar et al., 2020; Sheth et al., 2025); Hinglish-TOP covers task-oriented parsing in non-banking domains (Agarwal et al., 2023), and BANKING77 is English-only (Casanueva et al., 2020); MASSIVE is translated rather than code-mixed (FitzGerald et al., 2023). Realistic-noise studies report clean→noisy gaps of 2–4 points for intent tasks (Aliakbarzadeh et al., 2025); our messy tier matches this range, while our adversarial tier (-24 points) shows colloquial code-mix far exceeds "typo noise" (survey: Winata et al., 2023).

Cascades and routing. LLM cascades and routers reduce cost by sending easy inputs to cheap models (Chen et al., 2023; Ding et al., 2024; Ong et al., 2025; Varangot-Reille et al., 2025), and cascading can improve accuracy, not just cost (Varshney and Baral, 2022). Recent work, much of it in preprints, begins to name the wasted-escalation problem (Jitkrittum et al., 2023; Rabanser et al., 2025; Bouchard, 2026; Aggarwal et al., 2023); we contribute a measurement these works lack: escalation and rescue rates stratified by input noise tier. The closest system routes intent queries between a fine-tuned classifier and an LLM on uncertainty (Arora et al., 2024); it chooses *between two deciders*, whereas we show that restructuring the LLM's candidate list with a non-deciding ranker beats replacing either component.

Rankers as deciders vs. rankers. Pairwise cross-encoders have served as few-shot intent *deciders* in clean settings (Zhang et al., 2020). Our production result is explained by ranking theory: learning-to-rank objectives leave scores scale-uncalibrated, so relative order is trustworthy where absolute thresholds are not (Yan et al., 2022; Yu et al., 2025; Baral et al., 2026).

Augmentation pitfalls. LLM-based augmentation for intent classification is standard practice (Wei and Zou, 2019; Sahu et al., 2022, 2023), with known label-noise risks near class boundaries. Targeted Data Generation anticipates that failure-directed augmentation can harm other subgroups and engineers around it (He et al., 2023); counterfactual augmentation can suppress unperturbed robust features (Joshi and He, 2022); recent theory predicts synthetic data hurts when class imbalance is not the dominant error source (Ma and Zhang, 2026). We document the harm *materializing* in a production-style pipeline, and note that accumulation alongside real data (Gerstgrasser et al., 2024) did not prevent it. Robustness benchmarks for task-oriented dialogue (Liu et al., 2021; Peng et al., 2021) perturb inputs but do not ablate deployed pipelines layer-by-layer.

9 Conclusion

We asked where noise robustness lives in one production-style code-mixed banking NLU and measured an answer: it migrates. Repair rules carried it before encoder fine-tuning; afterward the encoder carries it, only for the pair it saw. Deep-noise decisions stay with a bounded LLM arbiter no substitute matched; the same small model, restricted to reranking, lifted recall and cut escalations. Layers in mature pipelines have *half-lives*, and they are measurable: factorial ablation crossed with training condition, plus a transfer probe, identifies which defenses are load-bearing and which are vestigial; remeasure per pair, register, and role. The protocol is system-agnostic; we hope others run it on theirs.

Limitations

Single domain (banking), whose heavy English loanwords likely flatter zero-shot transfer. Four evaluation sets across three code-mix pairs (Hinglish; two independent Tamil–English translations, one per register, so translator identity and register are confounded; Telugu–English at n=40); each transfer set is single-translator and small. The in-house benchmark has single-team labels, author-re-adjudicated (the normalizer finding is additionally replicated on public Hinglish-TOP). The reranker's end-to-end accuracy effect (+1.6 points) is not resolved to significance at n=330 by any test we report (bootstrap CI lower bound at zero; McNemar p=0.18); the mechanism measurements (shortlist recall and escalation reduction) are its firmest support. The arbiter rescue rate (§4.2) is measured indirectly, as the outcome delta between the baseline and –arbiter configurations, under a rest-of-pipeline invariance assumption. The arbiter runs a single prompt at temperature 0; we did not quantify prompt sensitivity, so part of its measured value may be prompt-specific. The arbiter and reranker findings rest on the self-authored benchmark alone; only the normalizer finding has a public-data replication.

Ethics Statement

All benchmark queries were authored by the author in the style of WhatsApp banking messages; none derive from real customer communications, and the customer-data platform used by the system is entirely synthetic (fictional profiles), so no personally identifiable information of any real individual appears in the system, the benchmark, or this paper. The Tamil and Telugu transfer sets were produced by consenting adult native-speaker collaborators (personal acquaintances of the author) who volunteered without monetary compensation. The task was bounded (reviewing or correcting two machine-generated candidates per query, ~200 rows per collaborator, a few hours in total), participation was declinable at any point, and no from-scratch translation was required; participation was voluntary and the collaborators will be acknowledged in the de-anonymized version. We acknowledge community norms favoring paid annotation and scope these sets accordingly as small validation probes rather than crowd-sourced resources. LLM API calls made during evaluation contained only synthetic queries. Misrouting a banking intent has real cost (executing the wrong consequential action), so the system requires explicit user confirmation before any high-risk action and abstains (CLARIFY/OOS) rather than guessing under low confidence; dialect and language coverage is uneven and untested pairs may be served worse, which is why we scope claims to the tuned pair. The system is a research build and serves no live banking customers; we make no deployment claims. An LLM-based coding and writing assistant was used throughout this work (for experiment tooling, analysis, and manuscript drafting) under author direction. The BankStress-330 queries are human-authored; the transfer-set candidate translations were machine-generated (llama-3.3-70b) and no machine-generated translation or label was accepted without native-speaker review; all benchmark labels were human-adjudicated. All experimental designs, data, and claims were specified and verified by the author, who takes full responsibility for the content.

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A Artifact Availability

Upon acceptance we release, under CC BY 4.0 (data) and Apache 2.0 (code): (1) BankStress-330, the tiered benchmark, with per-query authoring notes and the full adjudication record (raw and adjudicated labels); (2) the three native-validated transfer sets (Tamil $\times$2 registers, Telugu); (3) the ablation/reranker harness and result-verification scripts (documenting how each configuration was produced; runnable against the pipeline released at camera-ready); and (4) the fine-tuned cross-encoder reranker weights (camera-ready; at review time the released per-query rerank logs and the training script suffice to verify every reranker claim without the weights). The supplementary archive submitted with this paper is the authoritative artifact; an anonymized browsable mirror is also available at `https://anonymous.4open.science/r/nlu-wnut-artifacts-5B2D/`.

B Reproducibility and Compute Statement

All pipeline inference is CPU except the cross-encoder reranker and encoder fine-tuning, which used one NVIDIA RTX A4000 (16 GB), co-resident with an unrelated service throughout. Encoder adapter: LoRA rank 16, α=32, dropout 0.1 on query/value projections of multilingual-e5-base, triplet loss, batch 128, 6 epochs. Reranker fine-tune: 21,456 taxonomy-derived pairs, 1 epoch ($\sim$65 min). Arbiter: gpt-oss-20b served via a hosted API at temperature 0 with a fixed prompt and fixed decoding parameters; substitutes: Llama-3.1-8B-Instruct (same prompt and role) and bge-reranker-v2-m3 (568M). Encoder: intfloat/multilingual-e5-base. Full request parameters and per-call logs accompany the artifact where releasable. Wall-clock per 330-query evaluation run: $\sim$3 min with the arbiter, $<$1 min without; each 80-query transfer run $\sim$2 min. All nine selected headline statistics (flip rate, normalizer and arbiter discordances, holdout, shortlist recall, escalation rates, reranker pairwise, augmentation, and the Hinglish-TOP null) recompute from the released per-query records via `harness/verify_claims.py`; the bundle is a result-verification package (the pipeline source and reranker weights are released at camera-ready).